operators. But we need to select the right technologies for configuring infrastructure, optimising performance, and verifying the interoperability of new products or services in a multi-vendor environment.

Data centre operators are turning to next-generation transceivers to increase channel bandwidth to reach 400GE speeds. These transceivers guarantee quality and interoperability and reduce test time and cost. Upgrading from 100GE to 400GE requires a whole new set of testing for 400GE components and transceivers. The new transceiver technology must be thoroughly tested to comply with industry specifications to ensure seamless compatibility before deploying into the network.

Software-defined networking (SDN) has helped cloud and service providers to visualise, monitor and manage cloud-level traffic in real time. It has been adopted massively to accelerate the deployment of network services, save money and increase reliability.

Test equipment must be capable of supporting effective traffic simulation, testing components and systems at the new port rates, and standards compliance. All cables interconnecting the new port rates, whether optical fibre or direct-attached copper, and equipment like switches and routers need to be tested to guarantee proper operation.

Data centre testing and validation approach

The new network paradigm, coupled with rapidly changing hardware and protocols, puts special demands on data centre operators and the test equipment they use. As it is financially impractical to buy separate test gear for each speed, equipment must be multifunctional so that it can be used throughout the network as required. Automated tests increase the repeatability and reliability of results and simplify the process.

Next-generation test and measurement is part of an essential toolset for keeping the cloud and the edge working optimally with a zero-tolerance approach to latency and downtime. Relatively new forms of computing, such as serverless and edge, are taking some of the pressure off data centres, but these create other challenges instead. For instance, with growing bandwidth, the role of data centre interconnects (DCIs) in delivering on data demands with an optimally efficient architecture will become increasingly important.

Getting the most out of existing data centre infrastructure is a challenge that Internet content providers (ICPs) now face. Rigorous test and measurement can help ICPs manage change and speed network build-outs. For example, automated test and inspection practices upgrade their equipment to enable 100G and 400G speeds (and beyond). ICPs need a way of ensuring reliability. ICPs also need to ensure performance across the whole ecosystem with data processing becoming distributed and abstracted across hybrid environments.

The ANSI/TIA-942-A infrastructure standard defines the specific functional areas and contains appropriate information for the data centres, providing minimum recommendations for pathways and spaces, redundancy, cable management, environmental considerations and backbone, and horizontal cable media distances. It is similar to TIA-942-A and TIA-568 data centre standards like ISO/IEC 24764, which are applicable for information technology generic cabling systems for data centres, and ANSI/BICSI 002-2014. Data centre design and implementation best practices also outline various functional areas of the data centres that define equipment placement while allowing for scalability and reliability.

Measurement technologies for data centres

Virtual solutions by Spirent optimise and validate NFV infrastructure, virtual network functions, and network services by creating elastic test topologies to run on both network virtualisation and cloud platforms. Its OpenFlow solution validates and deploys SDN in a multitude of real-world scenarios of functional and performance testing. It helps to achieve compliance with OpenFlow standards and the performance of the controllers, switches and end-to-end open flow network designs.

With data centre cloud services, 4K video, and 5G mobile network services, many network operators are moving from 28Gb/s (NRZ) signalling to faster 56Gb/s (PAM4) electrical interfaces. Ixia's K400 QSFP-DD load module can help validate all technologies utilising 400GE-port fan-out cables for testing 50GE, 100GE, 200GE, or 400GE multi-speed or single-speed devices.

The measurement solutions include modules for evaluating 400G technologies used by cloud services and data centres. The MP2110A is Anritsu's solution for 25G/100G/200G/400G speeds, while the MP2100B targets 10G/40G applications. The MT1000A provides a solution for troubleshooting and installing DCI.

Video, cloud computing, and other bandwidth-intensive applications are forcing operators to rapidly increase network capacity. Network equipment manufacturers (NEMs) need best-in-class test equipment to support them in their transformations. EXFO offers 3D analytics and test methodologies to converged network testing for 100G and beyond, addressing the needs of NEMs for design, manufacturing, and verification. Thus it ensures scalability and resilience in data centre networks.

VIAVI collaborates with hyper-scale data centre operators, cloud service providers, ICPs, and those deploying DCI to optimise optical networks, reduce test time and latency, and ensure hundred per cent reliability that supports SLAs. It enables inspection of MPO connectors in twelve seconds, and of two 100G ports simultaneously. It is involved in all stages of hyper-scale optical testing, and tests high-speed networks up to 400G and beyond. END

By: Deepshikha Shukla

The author is a freelance technology journalist.

The article was originally published in the May 2020 issue of Electronics For You.